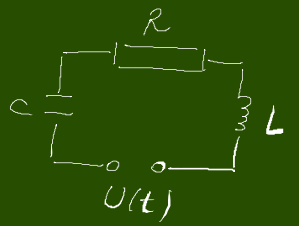
Ecuații diferențiale liniare de ordin superior cu coeficienți constanți

exemplu RLC serie



$U_C + U_R + U_L = U$

$U_R(t) = I(t) \cdot R$ (Ohm)

$U_L(t) = L \cdot I'(t)$ (L - autoinducție)

$U_C(t) = \frac{1}{C} Q(t)$

$I(t) = Q'(t)$

$$\frac{1}{C} Q(t) + R \cdot Q'(t) + L \cdot Q''(t) = U(t)$$

ec. dif. nec. $Q(t)$ de ordinul II

$Q(t), Q'(t), Q''(t)$

Definiție Ecuație diferențială liniară de ordinul n cu coeficienți constanți

$$x^{(n)} + a_1 x^{(n-1)} + \dots + a_{n-1} x' + a_n x = b(t) \qquad (1)$$

t variabilă

$x = x(t)$ fct. nec. $x', \dots, x^{(n)}$ derivatele fct. nec. $x(t)$

$a_1, \dots a_n \in \mathbb{R}$ $b : I \subset \mathbb{R} \longrightarrow \mathbb{R}$ fct. continuă

Condiții inițiale : $x(t_0) = x_0 \;,\; x'(t_0) = x_1 \;,\; \dots \;,\; x^{(n-1)}(t_0) = x_{n-1}$ (2)

$t_0 \in I \;,\; x_0, \dots, x_{n-1} \in \mathbb{R}$

Problema Cauchy

$$\begin{cases} x^{(n)} + a_1 x^{(n-1)} + \dots + a_{n-1} x' + a_n x = b(t) \\ x(t_0) = x_0 \\ \vdots \\ x^{(n-1)}(t_0) = x_{n-1} \end{cases}$$

$b(t) = 0 \Rightarrow$ ec. diff. omogenă

$b(t) \neq 0 \Rightarrow$ ec. diff. neomogenă.

Observație

Fie $x(t)$ o soluție a ec. diff. (1)

notăm

$$\begin{aligned} y_1(t) &= x(t) \\ y_2(t) &= x'(t) \\ &\vdots \\ y_n(t) &= x^{(n-1)}(t) \end{aligned}$$

$$\Rightarrow \begin{cases} y_1' = y_2 \\ y_2' = y_3 \\ \vdots \\ y_{n-1}' = y_n \\ y_n' = -a_n y_1 - a_{n-1} y_2 - \dots - a_1 y_n + b(t) \end{cases} \quad (3)$$

$\overset{?}{\Leftarrow}$

Fie $\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$ o soluție a sistemului de ec. dif. (3)

dacă notez $x(t) = y_1(t) \Rightarrow x'(t) = y_1'(t) = y_2(t)$

$x''(t) = y_3(t)$

$\dots$

$x^{(n-1)}(t) = y_n(t)$

$x^{(n)}(t) = y_n'(t) = -a_n y_1(t) - a_{n-1} y_2(t)$

$- \dots - a_1 y_n(t) + b(t)$

$= -a_n x(t) - a_{n-1} x'(t) \dots - a_1 x^{(n-1)}(t)$

$+ b(t)$

$\Rightarrow x(t)$ este o soluție a ec. dif. de ordinul n (1)

Obs. că $x(t)$ satisface condiția inițială (2) $\Leftrightarrow$

$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$ satisface condiția $\begin{cases} y_1(t_0) = x_0 \\ y_2(t_0) = x_1 \\ \vdots \\ y_n(t_0) = x_{n-1} \end{cases}$ (4)

$\Rightarrow$

Proposizione 1) Problema Cauchy $\begin{cases} x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x = b(t) \\ x(t_0) = x_0 \quad \dots \quad x^{(n-1)}(t_0) = x_{n-1} \end{cases}$

ha sol. unica $\forall\, t_0 \in I$, $x_0, \dots, x_{n-1} \in \mathbb{R}$.

ec. diff. omogenea

Conseguenza di $b(t) = 0 \Rightarrow$ allora pb. Cauchy $\begin{cases} x^{(n)} + \dots + a_n x = 0 \\ x(t_0) = 0 \dots x^{(n-1)}(t_0) = 0 \end{cases}$

ha sol. unica $x(t) = 0 \quad \forall\, t \in \mathbb{R}$

2) $b(t) = 0$ (ec. diff. omogenea)

$S_0 = \{ x : \mathbb{R} \to \mathbb{R} \mid x$ deriv. n volte e $\underbrace{x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x = 0}_{(S)} \}$

$S_0 = \{ Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} : \mathbb{R} \to \mathbb{R}^n \mid Y$ fct. deriv. e $Y' = AY \}$

dove $A = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & & & & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_n & -a_{n-1} & & \dots & -a_1 \end{pmatrix}$

S_0 sp. vettoriale di dimensione n

$\Rightarrow$ $\mathcal{S}_0$ sp. vectorial peste $\mathbb{R}$ $\Rightarrow$ $\mathcal{S}_0 \underset{\text{izomorf}}{\simeq} S_0$ $\left(\Rightarrow \dim_{\mathbb{R}} \mathcal{S}_0 = n\right)$

3) $b(t) \neq 0$ (ec. dif. neomogenă)

$\mathcal{S}_{n0} = \{ x : I \longrightarrow \mathbb{R} \mid x \text{ derivabile de } n \text{ ori și } x \text{ verif. ec. dif. (1)} \}$

Fie $x_p \in \mathcal{S}_{n0}$ o sol. particulară.

atunci $x \in \mathcal{S}_{n0} \iff x - x_p \in \mathcal{S}_0$

$\left(\mathcal{S}_{n0} = \mathcal{S}_0 + \{x_p\} \right)$

$x = x_0 + x_p$.

Consec. $\dim_{\mathbb{R}} \mathcal{S}_0 = n$ $\{x_1(t) \dots x_n(t)\}$ bază în $\mathcal{S}_0$

(sistem fundamental de soluții)

$\Rightarrow$ sol. generală a ec. dif. (1) este

$x(t) = C_1 x_1(t) + \dots + C_n x_n(t)$ unde $C_1 \dots C_n \in \mathbb{R}$

$n = 2$ **Ecuația diferențială liniară de ordinul 2 omogenă**

(6) $\boxed{x'' + ax' + bx = 0}$ $\quad a, b \in \mathbb{R}$ $\quad \begin{cases} y_1' = y_2 \\ y_2' = -by_1 - ay_2 \end{cases}$

$n = 1$ $\quad x' + ax = 0 \quad \Rightarrow x' = -ax \Rightarrow x(t) = Ce^{-at}, C \in \mathbb{R}$

Caut o sol. a ec. dif. (6) de forma $x(t) = e^{\lambda t}$, $\lambda \in \mathbb{R}$

$x'(t) = \lambda e^{\lambda t}$

$x''(t) = \lambda^2 e^{\lambda t}$

$A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}$

$P_A(X) = X^2 + aX + b$

$\Rightarrow \lambda^2 e^{\lambda t} + a\lambda e^{\lambda t} + b e^{\lambda t} = 0$

$\boxed{\lambda^2 + a\lambda + b = 0}$ ecuația caracteristică asociată ec. dif. (6)

$\lambda_1, \lambda_2 \in \mathbb{R}$, $\lambda_1 \neq \lambda_2$

$\lambda_1 = \lambda_2 \in \mathbb{R}$

$\lambda_1, \lambda_2 \in \mathbb{C} - \mathbb{R}$, $\lambda_2 = \overline{\lambda_1}$

I $\quad \lambda^2 + a\lambda + b = 0 \quad \begin{cases} \lambda_1 \in \mathbb{R} \\ \lambda_2 \in \mathbb{R} \end{cases} \quad \lambda_1 \neq \lambda_2$

$\Rightarrow$ $x_1(t) = e^{\lambda_1 t}$, $x_2(t) = e^{\lambda_2 t}$ sol. pt ec. dif. (6)

Vectorii ec. x_1, x_2 f.d. liniar independente $\begin{cases} \text{dacă} \\ \mathcal{S}_0 \simeq \mathbb{S}_0 \end{cases}$

x_1, x_2 lin. indep. $\Leftrightarrow \begin{pmatrix} x_1 \\ x_1' \end{pmatrix}, \begin{pmatrix} x_2 \\ x_2' \end{pmatrix} \in \mathbb{S}_0$ lin. indep.

$\Leftrightarrow \begin{pmatrix} x_1(t_0) \\ x_1'(t_0) \end{pmatrix}, \begin{pmatrix} x_2(t_0) \\ x_2'(t_0) \end{pmatrix}$ lin. indep. în $\mathbb{R}^2$ $(\forall t_0 \in \mathbb{R})$

Fie $t_0 = 0$

$\begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix}, \begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix} \in \mathbb{R}^2$ lin. indep. $(\Leftrightarrow \lambda_1 \neq \lambda_2)$

$\Rightarrow$ x_1, x_2 f.d. lin. indep $\Rightarrow$ x_1, x_2 bază în $\mathcal{S}_0$

$\Rightarrow$ sol. generală a ec. dif. $x'' + ax' + bx = 0$ este

$x(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$, $C_1, C_2 \in \mathbb{R}$.

Exemplu. $x'' - 2x' - 8x = 0$, $x(0) = 5$, $x'(0) = 2$

ec. caracteristică $\lambda^2 - 2\lambda - 8 = 0$

$(\lambda + 2)(\lambda - 4) = 0$

$\lambda_1 = 4 \Rightarrow x_1(t) = e^{\lambda_1 t} = e^{4t}$

$\lambda_2 = -2 \Rightarrow x_2(t) = e^{\lambda_2 t} = e^{-2t}$

$\Rightarrow$ sol. gen. a ec. dif $x'' - 2x' - 8x = 0$ este $x(t) = C_1 x_1(t) + C_2 x_2(t)$

$= C_1 e^{4t} + C_2 e^{-2t}$

$x(0) = 5 \Rightarrow C_1 + C_2 = 5 \quad / \cdot 2$

$x'(0) = 2 \Rightarrow 4C_1 - 2C_2 = 2$

$6C_1 = 12 \quad C_1 = 2$

$C_2 = 3$

$x'(t) = 4C_1 e^{4t} - 2C_2 e^{-2t}$

$\Rightarrow x(t) = 2e^{4t} + 3e^{-2t}$.

II $\lambda_1 = \lambda_2 \in \mathbb{R}$ not. $\lambda_1 = \lambda_2 = \lambda$ nod. a ec. caract. $\lambda^2 + a\lambda + b = 0$

$\Rightarrow x_1(t) = e^{\lambda t}$ sol. a ec. dif $x'' + ax' + b = 0$

$x_2(t) = te^{\lambda t}$ x_2 sol. a ec. dif

x_1, x_2 sist. indep.

$x_2(t) = te^{\lambda t}$

$x_2'(t) = e^{\lambda t} + t\lambda e^{\lambda t}$

$x_2''(t) = 2\lambda e^{\lambda t} + t\lambda^2 e^{\lambda t}$

$\Rightarrow e^{\lambda t}(2\lambda + t\lambda^2) + ae^{\lambda t}(1 + t\lambda) + bte^{\lambda t} \stackrel{?}{=} 0$

$t(\underbrace{\lambda^2 + a\lambda + b}_{0}) + (\underbrace{2\lambda + a}_{0}) \stackrel{?}{=} 0$ (Viète)

$\Rightarrow x_2(t) = te^{\lambda t}$ sol. nec. dif.

Testăm liniar indep: x_1, x_2 lin. indep. $\Leftrightarrow \begin{pmatrix} x_1 \\ x_1' \end{pmatrix}, \begin{pmatrix} x_2 \\ x_2' \end{pmatrix}$ lin. indep.

$\Leftrightarrow \begin{pmatrix} x_1(0) \\ x_1'(0) \end{pmatrix}, \begin{pmatrix} x_2(0) \\ x_2'(0) \end{pmatrix} \in \mathbb{R}^2$ lin. indep.

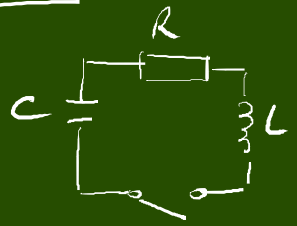
$\begin{pmatrix} 1 \\ \lambda \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $\quad \begin{vmatrix} 1 & 0 \\ \lambda & 1 \end{vmatrix} = 1 \neq 0$

$\Rightarrow$ lin. indep.

$\Rightarrow x_1(t) = e^{\lambda t}$, $x_2(t) = te^{\lambda t}$ sistem fundam. de sol. pt $x'' + ax' + bx = 0$

$\Rightarrow$ sol. gen. $x(t) = C_1 x_1(t) + C_2 x_2(t) = C_1 e^{\lambda t} + C_2 te^{\lambda t}$ $\quad C_1, C_2 \in \mathbb{R}$.

Exemplu. RLC serie $\quad L \cdot Q''(t) + R \cdot Q'(t) + \frac{1}{C} Q(t) = 0$



$L = 1\,H, \quad R = 2\,\Omega, \quad C = 1\,F$

$Q(0) = 2\,C \qquad I(0) = 1\,A.$

$\lim_{t \to \infty} Q(t) = ? \qquad \lim_{t \to \infty} I(t) = ?$

$Q'' + 2Q' + Q = 0$

$\lambda^2 + 2\lambda + 1 = 0 \Rightarrow \lambda = -1 \begin{cases} x_1(t) = e^{\lambda t} = e^{-t} \\ x_2(t) = te^{\lambda t} = te^{-t} \end{cases}$

$\Rightarrow Q(t) = C_1 x_1(t) + C_2 x_2(t) = C_1 e^{-t} + C_2 te^{-t}. \quad C_1, C_2 \in \mathbb{R}.$

$Q(0) = 2 \Rightarrow C_1 = 2$

$I(0) = Q'(0) = 1 \Rightarrow -C_1 + C_2 = 1 \Rightarrow C_2 = 3$

$I(t) = Q'(t) = -C_1 e^{-t} + C_2 e^{-t} - C_2 te^{-t}$

$\Rightarrow Q(t) = \underline{2e^{-t} + 3te^{-t}} = e^{-t}(2 + 3t) \xrightarrow{t \to \infty} 0$

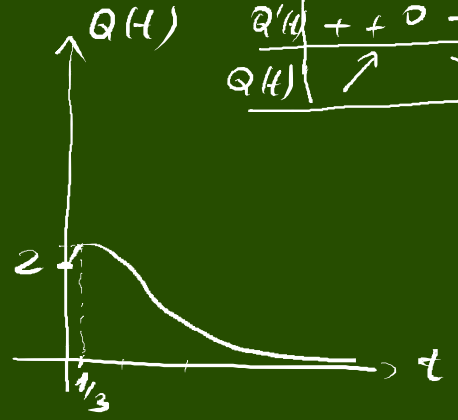
$I(t) = e^{-t}(1-3t) \xrightarrow{t\to\infty} 0$

$Q'(t) = 0 \iff t = \frac{1}{3}$

$Q(\frac{1}{3}) = 3e^{-\frac{1}{3}} \approx 0{,}716 \cdot 3$

$e^{-\frac{1}{3}}\left(2 + 3\cdot\left(\frac{1}{3}\right)\right) = 3e^{-1/3} \approx 2{,}1 > 2$

t	0	$\frac{1}{3}$	∞
$Q'(t)$	$+$	0	$-$
$Q(t)$	$\nearrow$		$\searrow$



III $\quad \lambda_1, \lambda_2 \in \mathbb{C} - \mathbb{R}$, $\lambda_2 = \bar{\lambda}_1$ $\qquad$ Not: $\lambda_1 = \lambda = \alpha + i\beta$, $\beta \neq 0$.
$\lambda_2 = \bar{\lambda} = \alpha + i\beta$

$\Rightarrow \quad z(t) = e^{\lambda t} = e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos\beta t + i \sin\beta t)$ $\qquad$ Sol. complexă a ec. dif $x'' + ax' + bx = 0$.

$$\begin{cases} x_1(t) = \operatorname{Re} z(t) = e^{\alpha t}\cos\beta t \\ x_2(t) = \operatorname{Im} z(t) = e^{\alpha t}\sin\beta t \end{cases}$$

x_1, x_2 sol. $\longrightarrow$ x_1, x_2 liniar independente.

$z'' + az' + bz = 0 \qquad z = x_1 + ix_2$

$\Rightarrow x_1'' + ix_2'' + a(x_1' + ix_2') + b(x_1 + ix_2) = 0 \quad \left.\begin{matrix} \\ \end{matrix}\right\} \Rightarrow \begin{matrix} x_1'' + ax_1' + bx_1 = 0 \\ x_2'' + ax_2' + bx_2 = 0 \end{matrix}$

$a, b \in \mathbb{R}$

$\Rightarrow x_1, x_2$ sol. ale ec. dif $x'' + ax' + bx = 0$

x_1, x_2 liniar indep. $\Leftrightarrow \begin{pmatrix} x_1 \\ x_1' \end{pmatrix}, \begin{pmatrix} x_2 \\ x_2' \end{pmatrix}$ liniar indep $\Leftrightarrow \begin{pmatrix} x_1(0) \\ x_1'(0) \end{pmatrix}, \begin{pmatrix} x_2(0) \\ x_2'(0) \end{pmatrix} \in \mathbb{R}^2$ liniar indep.

$\begin{pmatrix} x_1(0) \\ x_1'(0) \end{pmatrix} = \begin{pmatrix} 1 \\ \alpha \end{pmatrix} \qquad \begin{pmatrix} x_2(0) \\ x_2'(0) \end{pmatrix} = \begin{pmatrix} 0 \\ \beta \end{pmatrix}$

$x_1'(t) = \alpha e^{\alpha t} \cos \beta t + e^{\alpha t}(-\beta \sin \beta t)$

$x_2'(t) = \alpha e^{\alpha t} \sin \beta t + e^{\alpha t} \beta \cos \beta t$

$\begin{vmatrix} 1 & 0 \\ \alpha & \beta \end{vmatrix} = \beta \neq 0 \Rightarrow x_1, x_2$ fol. liniar indep.

$\Rightarrow x_1(t) = e^{\alpha t} \cos \beta t, \quad x_2(t) = e^{\alpha t} \sin \beta t$ S.F.S. fundam. de sol.

pt. ec. dif. $x'' + ax' + bx = 0$

$\Rightarrow$ sol. generală $x(t) = C_1 x_1(t) + C_2 x_2(t) = C_1 e^{\alpha t} \cos \beta t + C_2 e^{\alpha t} \sin \beta t.$

$C_1, C_2 \in \mathbb{R}$.

Exemplu. $x'' + 6x' + 13x = 0 \quad (x(0) = 0 \quad x'(0) = 0)$

$\Downarrow$

$x(t) = 0 \quad \forall t \in \mathbb{R}$

$x(0) = 1, \quad x'(0) = -1$

$\lambda^2 + 6\lambda + 13 = 0 \qquad (\lambda + 3)^2 + 4 = 0 \qquad \lambda_{1,2} = -3 \pm 2i$

$-3 + 2i \quad \alpha = -3$

$\beta = 2$

$\Rightarrow x_1(t) = e^{\alpha t} \cos \beta t = e^{-3t} \cos 2t \qquad x_2(t) = e^{\alpha t} \sin \beta t = e^{-3t} \sin 2t$

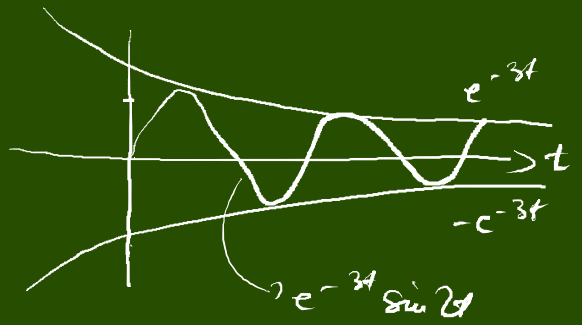
$\Rightarrow x(t) = C_1 e^{-3t} \cos 2t + C_2 e^{-3t} \sin 2t.$

$x(0) = 1 \Rightarrow C_1 = 1$

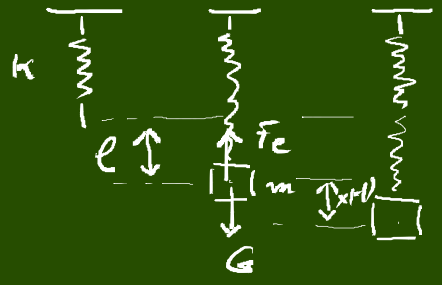
$x'(0) = -1 \Rightarrow -3C_1 + 2C_2 = -1 \Rightarrow C_2 = 1$

$x'(t) = -3C_1 e^{-3t} \cos 2t - 2C_1 e^{-3t} \sin 2t - 3C_2 e^{-3t} \sin 2t + 2C_2 e^{-3t} \cos 2t$

$\Rightarrow \underbrace{x(t)}_{\to 1} = \underbrace{e^{-3t}}_{\to 0} \cos 2t + \underbrace{e^{-3t}}_{\to 0} \sin 2t \xrightarrow{t \to \infty} 0$



Exemplu



$mg = k\ell$

$$ma = mg - k(x(t)+\ell) - F_f$$

$$F_f = \mu v$$

$$v(t) = x'(t)$$

$$a(t) = x''(t)$$

$$m \cdot x'' = mg - kx - k\ell - \mu \cdot x'$$

$$x'' + \underbrace{\frac{\mu}{m}}_{2a} x' + \underbrace{\frac{k}{m}}_{\omega^2} x = 0$$

$$\lambda^2 + 2a\lambda + \omega^2 = 0$$

$\Delta = 4a^2 - 4\omega^2$ $\quad$ I $\quad \Delta < 0 \quad 0 \le a < \omega$

I.1. $a = 0$ (fără frecare)

$\Delta = -4\omega^2 < 0 \qquad \lambda_{1,2} = \pm i\omega \Rightarrow x_1(t) = \cos \omega t$

$\lambda^2 + \omega^2 = 0 \qquad x_2(t) = \sin \omega t$

$\Rightarrow x(t) = C_1 \cos \omega t + C_2 \sin \omega t$ mișcare oscilatorie

periodică cu perioada $\frac{2\pi}{\omega}$

frecvență $\frac{\omega}{2\pi}$

pulsație ω

amplitudinea mișcării ?

$x(t) = A \sin(\omega t + \varphi_0)$

↳ amplitudinea. ↳ faza inițială